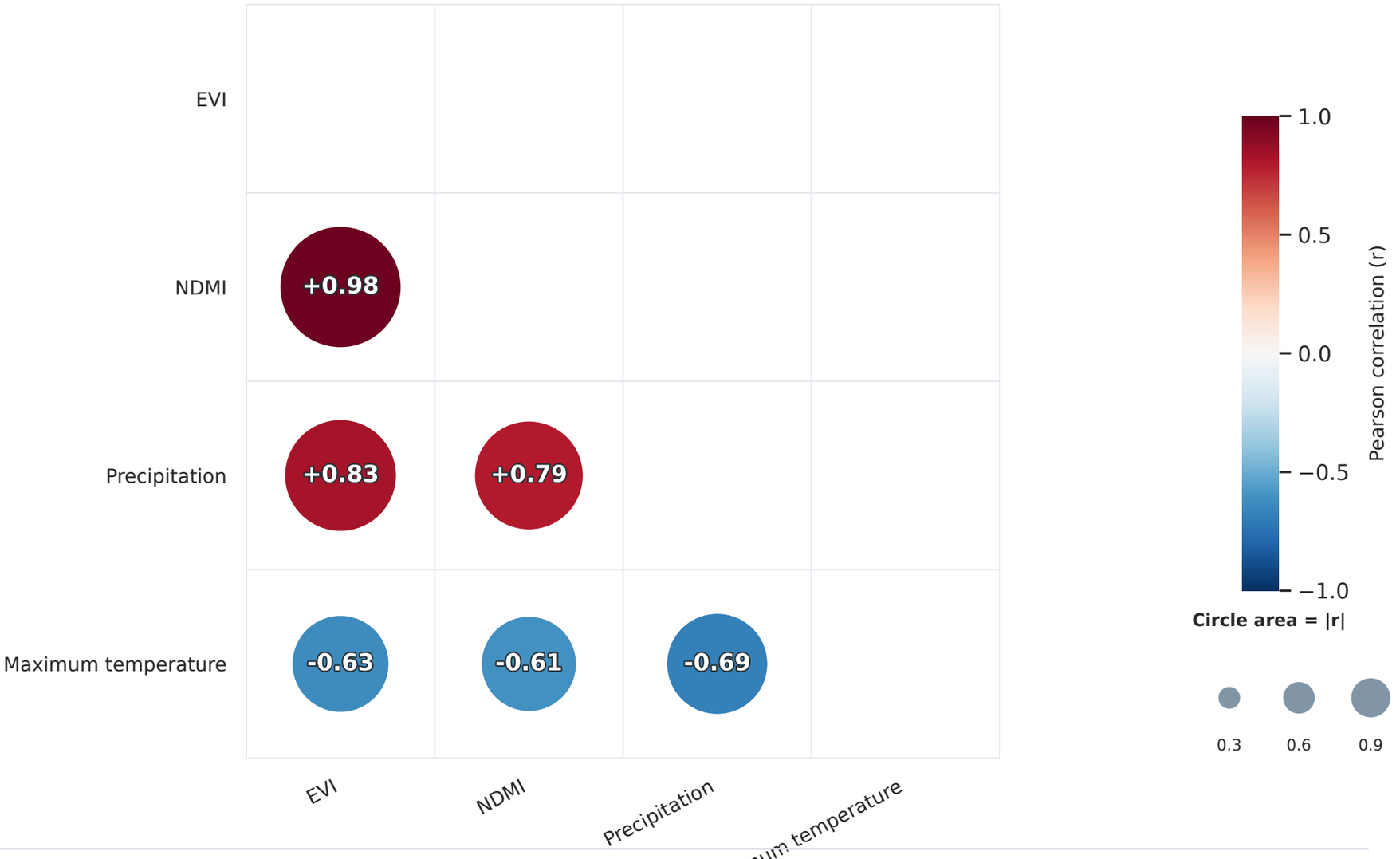
VEGETATION–CLIMATE CORRELATION STRUCTURE

Environmental Data Cube for Oglala Lakota

Ate Makoce | May–September 2000–2023 | 24 annual observations

Pairwise Pearson correlations



RESULTS INTERPRETATION

The strongest positive relationship occurred between EVI and NDMI (r = +0.98). The strongest negative relationship occurred between Precipitation and Maximum temperature (r = -0.69).

Interpretive limitation: Pearson correlations describe linear associations, not causation. Annual observations may exhibit temporal dependence, and vegetation and climate data may have different spatial footprints.

Source: Annual EVI and NDMI validation series; gridMET precipitation and maximum temperature. Growing-season aggregates, 2000–2023.